
Modality-Conditioned Safety Representations for Vision-Language Models

Mid-Project Report

Abstract

Vision-Language Models (VLMs) exhibit a subtle safety failure: given a *safe image* paired with a *safe text prompt*, the model sometimes produces an unsafe response—a phenomenon we term SSU (Safe-image, Safe-text $\rightarrow$ Unsafe output). We argue this failure calls for **modality-conditioned safety representations**: safety boundaries that adapt based on how much each modality contributes to the input. As a first step, we apply the SHIFDC diagnostic [9] to LLaVA-1.5-7B on HoliSafe-Bench [3] (682 SSS + 718 SSU samples), computing a per-layer modality shift $m^l = x_{\text{vi}}^l - x_{\text{tt}}^l$ and projecting it onto a PCA-derived safety direction s^l . We find significant SSS/SSU divergence in 25 of 32 layers ($p < 0.001$), with the largest gap at layer 31 ($p = 5.5 \times 10^{-30}$). A text-only sanity check reveals that SSU prompts are already semantically closer to unsafe content before image integration, suggesting the failure is partly prompt-driven. Building on these diagnostic findings, we are developing a modality-conditioned correction mechanism guided by attention allocation signals.

1 Introduction

Vision-Language Models (VLMs) extend large language models with visual perception, enabling reasoning over image-text pairs [?]. Despite strong performance on downstream tasks, VLMs introduce safety vulnerabilities qualitatively distinct from their text-only counterparts. Prior work has shown that safety-relevant concepts in large language models are encoded as linear directions in activation space [1], enabling effective safety classification through geometric methods. However, VLMs face a more complex challenge: visual inputs induce systematic shifts in these safety representations [9], causing the model to perceive certain multimodal inputs as *safer* than equivalent text-only queries.

A striking manifestation of this distortion is the **SSU phenomenon**, identified in HoliSafe-Bench [3]: given a *safe image* and a *safe text prompt*, the model nonetheless produces an unsafe response. Neither modality alone triggers the failure, and text-only safety mechanisms (RLHF, content classifiers) are blind to it.

We argue that effective VLM safety detection requires **modality-conditioned safety representations**—boundaries that adapt based on the dominant modality contributing to a given input. Our project proceeds in two phases:

1. **Diagnostic phase (complete):** Characterize SSS/SSU activation differences layer-by-layer using the SHIFDC framework.
2. **Intervention phase (in progress):** Learn modality-conditioned safety subspaces guided by the diagnostic findings.

2 Related Work

Linear representation of safety. Arditi et al. [1] demonstrated that refusal behavior in LLMs is mediated by a *single* linear direction in activation space, and that ablating this direction reliably suppresses refusal across diverse harmful inputs. This geometric structure underpins our use of a single per-layer safety direction s^l .

Modality-induced safety distortion. SHIFTDC [9] shows that visual inputs cause consistent activation shifts that distort safety perception: multimodal inputs appear safer than text-only equivalents in hidden-state geometry even when semantic content is held fixed. The authors propose a global projection-based correction; a key limitation is that this correction is modality-agnostic. Our work addresses this by conditioning safety boundaries on the dominant modality.

Attention imbalance in VLMs. Recent mechanistic studies have observed that text tokens attend to image tokens far less than their proportion of the input would suggest, contributing to safety vulnerabilities. This attention imbalance motivates using the attention allocation toward image tokens as a lightweight modality conditioning signal in our Phase 2 classifier.

Benchmarks. HoliSafe-Bench [3] provides per-modality safety labels across five image-text combinations, making it uniquely suited for SSS/SSU differential analysis. MM-SafetyBench [6] and JailBreakV-28K [?] target adversarial attacks.

3 Technical Approach

3.1 Phase 1: Activation Extraction and Safety Direction

For each sample in HoliSafe-Bench we extract hidden states at every transformer layer via two forward passes through LLaVA-1.5-7B:

- **VL activations** $x_{\text{vl}}^l \in \mathbb{R}^d$: last-token hidden state from the full multimodal input (image + text prompt).
- **TT activations** $x_{\text{tt}}^l \in \mathbb{R}^d$: last-token hidden state from a text-only surrogate (generated image caption + text prompt).

The **modality shift** $m^l = x_{\text{vl}}^l - x_{\text{tt}}^l$ isolates the contribution of visual integration. Captions are generated by LLaVA-1.5-7B (batched, max_new_tokens=100). All activations are stored via an on-disk ActivationCache for reproducibility.

PCA-based safety direction. We compute s^l from two reference datasets: LLaVA-Instruct-80k (safe) and MM-SafetyBench [6] (unsafe). Rather than the simple mean difference (which confounds safety with semantic content of the reference corpora), we use a PCA-based steering vector [9]:

1. Collect $H^+ \in \mathbb{R}^{N^+ \times d}$ (safe) and $H^- \in \mathbb{R}^{N^- \times d}$ (unsafe).
2. Jointly mean-center: $\mu = \frac{1}{2}(\bar{H}^+ + \bar{H}^-)$, $M = [H^+ - \mu; H^- - \mu]$.
3. s^l = first right singular vector of M , sign-flipped to align with $\bar{H}^+ - \bar{H}^-$ (unsafe $\rightarrow$ safe).

Shift projection. For each sample i and layer l :

$$\text{cosine}_i^l = \cos(m_i^l, s^l), \quad \text{proj}_i^l = \frac{m_i^l \cdot s^l}{\|s^l\|^2}. \quad (1)$$

Positive values indicate the modality shift is directed toward the *safe* side. We compare SSS vs. SSU groups per layer via Welch’s t -test.

3.2 Phase 2: Modality-Conditioned Safety Classifier (In Progress)

Building on the diagnostic findings, we are developing a modality-conditioned safety boundary. We use attention allocation as a lightweight conditioning signal. Let $\alpha_{\text{img}}^{(l)}$ be the fraction of attention

Table 1: Per-layer SHIFTDc cosine similarity results (selected layers). Positive gap = SSS > SSU. p -values from Welch’s t -test (two-sided).

Layer	SSS cosine	SSU cosine	Gap	p -value
1	0.075	0.072	−0.002	0.188 (n.s.)
4	0.452	0.456	−0.004	6.8×10^{-4}
8	0.260	0.283	−0.022	3.0×10^{-20}
12	0.041	0.068	−0.027	3.5×10^{-15}
20	0.057	0.039	+0.018	1.0×10^{-9}
31	0.092	0.055	+0.037	5.5×10^{-30}

mass on image tokens at layer l . Define the modality score:

$$s_{\text{mod}}(\mathbf{x}) = \frac{1}{L} \sum_{l=1}^L \alpha_{\text{img}}^{(l)}, \quad (2)$$

and interpolate between two learned projection matrices:

$$\mathbf{P}(\mathbf{x}) = (1 - s_{\text{mod}}) \mathbf{P}_{\text{txt}} + s_{\text{mod}} \mathbf{P}_{\text{img}}. \quad (3)$$

Safety is classified as $f(\mathbf{x}) = \text{sign}(\mathbf{P}(\mathbf{x}) h^{l^*} \cdot \mathbf{w} + b)$, where the critical layer l^* is selected via the diagnostic results (Section 4). We jointly learn $\mathbf{P}_{\text{txt}}$ and $\mathbf{P}_{\text{img}}$ with a contrastive objective:

$$\mathcal{L} = \mathcal{L}_{\text{contrastive}} + \lambda \|\mathbf{P}_{\text{txt}} - \mathbf{P}_{\text{img}}\|_F, \quad (4)$$

where the regularizer encourages the two projections to differ meaningfully.

4 Progress & Preliminary Results

Setup. Model: LLaVA-1.5-7B [?] ($d = 4096$, 32 transformer layers). Dataset: HoliSafe-Bench, filtered to **682 SSS + 718 SSU** samples. Reference datasets for safety direction: LLaVA-Instruct-80k (safe) and MM-SafetyBench (unsafe). Phase 1 is fully implemented as a reproducible pipeline (run_shiftdc.sh) with modular GPU extraction and CPU analysis stages.

4.1 Main SHIFTDc Results

Table 1 shows representative results across 32 layers. SSS and SSU groups show statistically significant differences ($p < 0.001$) in **25 of 32 layers**. Two regimes emerge:

Early-to-mid layers (4–14): SSU samples exhibit a *larger* positive modality shift toward the safe direction. This may indicate that the model attempts to regulate its internal representation as the unsafe potential of these samples is not yet fully represented.

Late layers (20–31): The pattern reverses—SSS samples show a larger safe-direction projection. The maximum gap at **layer 31** (SSS = 0.092, SSU = 0.055, $p = 5.5 \times 10^{-30}$) indicates that SSU final representations sit closer to the unsafe side of s^l , consistent with the unsafe output. This identifies layer 31 as a strong candidate for l^* in the Phase 2 classifier.

4.2 Sanity Check: Text-Only Baseline

This experiment tests whether the SSS/SSU difference is caused by image integration or by pre-existing semantic differences in the prompts.

Finding: In **28 of 32 layers**, SSU caption-only (TT) activations project more strongly onto the *unsafe* side of s^l than SSS TT activations. For example, at Layer 22: SSS projection = −5.09, SSU projection = −13.82.

This indicates that SSU prompts are semantically closer to unsafe content *before* any image is processed. The larger positive modality shifts seen in early layers (Section 4.1) likely reflect this baseline difference rather than purely image-induced distortion. This finding motivates the modality-conditioning approach: a safety boundary that ignores the text-level baseline risks either over-triggering on benign multimodal inputs or missing failures driven by prompt semantics.

4.3 Experiment A: Effective Rank of Modality Shift Space

The modality shift occupies a **low-dimensional subspace** relative to $d = 4096$. At layer 2 with cumulative variance threshold $\tau = 0.9$, effective rank $\approx 88\text{--}96$ for both SSS and SSU. Crucially, the two groups show nearly identical rank profiles across all layers, indicating the SSS/SSU difference is in the *direction* of the shift, not its dimensionality. The low rank ($\ll d$) confirms that safety-relevant distortions are structured and suggests that a low-rank projection $\mathbf{P}$ in Phase 2 is well-motivated.

5 Challenges & Risks

Text-level confound in SSU samples. The sanity check shows that SSU prompts are semantically closer to unsafe content in text space. Attributing the failure purely to image integration would be incorrect, and disentangling the two effects requires controlled ablations not yet completed.

Layer selection sensitivity. Safety directions vary substantially across transformer layers. The diagnostic analysis identifies layer 31 as the most discriminative, but selecting l^* robustly for the Phase 2 classifier requires careful cross-layer validation to avoid overfitting.

Semantic confound in reference datasets. The safety direction s^l is computed from LLaVA-Instruct vs. MM-SafetyBench, which differ not only in safety-relevance but also in topic distribution and writing style. The PCA method reduces this confound but does not eliminate it. We plan to introduce a paired text-only dataset (safeunsafe) with more controlled distributions.

Generalization of linear interpolation. The attention-based modality conditioning assumes a smooth spectrum between text-dominant and image-dominant regimes. For SSU inputs—where both modalities are safe individually—the conditioning signal may be ambiguous, potentially requiring a non-linear conditioning approach.

Compute constraints. Running two forward passes per sample across 32 layers on a 7B-parameter model is memory-intensive. Extending to larger models (e.g., LLaVA-1.5-13B) or secondary benchmarks (MM-SafetyBench, JailBreakV-28K) requires additional GPU time.

6 Next Steps & Timeline

Table 2: Remaining project milestones.

Dates	Milestone
Mar 31 – Apr 6	Complete follow-up experiments (B: safety decomposition, C: subspace overlap, D: category analysis); determine whether SSU distortion is concentrated in s^l or spread across multiple dimensions.
Apr 7 – Apr 13	Integrate paired safeunsafe reference dataset; recompute s^l and compare to LLaVA-Instruct/MM-SafetyBench baseline. Begin Phase 2 implementation: extract attention allocation signals and implement interpolated projection classifier.
Apr 14 – Apr 20	Evaluate Phase 2 classifier on HoliSafe-Bench SSU vs. SSS; compare against the ShiftDC [9] baseline. Ablate layer choice, λ , and linear vs. nonlinear conditioning.
Apr 21 – Apr 27	Extend analysis to LLaVA-1.5-13B to test generality; produce PCA visualizations and error analysis for final report.
Apr 28 – May 4	Final report revisions and submission.

The core diagnostic pipeline (Phase 1) is fully implemented and reproducible. The primary remaining work is: (1) completing follow-up experiments B–D to characterize the geometry of safety distortion; (2) addressing the semantic confound through a cleaner reference dataset; (3) implementing and evaluating the Phase 2 modality-conditioned classifier; and (4) validating findings on a second model.

References

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